

Electrical Components

ESP32-S3 Feather

<https://learn.adafruit.com/adafruit-esp32-s3-feather/overview>
<https://learn.adafruit.com/adafruit-esp32-s3-feather/downloads>

Accelerometer

[MPU6050 Gyrometer And Accelerometer Module - Datasheet Hub](#)

<https://www.youtube.com/watch?v=KuekQ-m9xpw>
<https://forum.digikey.com/t/accelerometer-types-applications-and-selection-tips/44520>

Magnetometer

<https://www.amazon.com/HiLetgo-GY-271-QMC5883L-Compass-Magnetometer/dp/B008V9S64>
<https://handsontec.com/dataspecs/sensor/GY-271%20HMC5883L.pdf>
<https://www.makeuseof.com/arduino-gy-271-hmc5883l-compass-magnetometer/>

RGB sensor

<https://protosupplies.com/product/tcs34725-rgb-color-sensor-module/>
https://www.rohm.com/electronics-basics/sensors/sensor_what7
<https://www.utmel.com/blog/categories/sensors/basic-introduction-to-color-sensor#2>
<https://www.makerguides.com/tcs34725-rgb-color-sensor-with-arduino/>

Power Management Module

<https://wiki.seeedstudio.cozm/Lipo-Rider-Plus/>

Motor controller

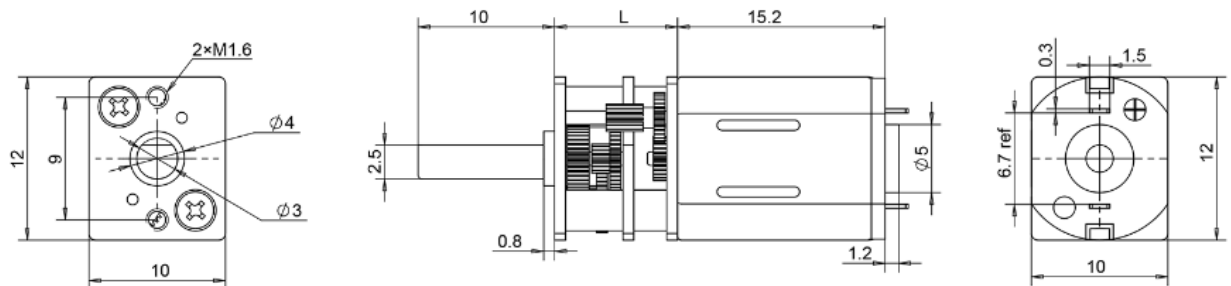
<https://cdn-learn.adafruit.com/downloads/pdf/adafruit-drv8833-dc-stepper-motor-driver-breakout-board.pdf>

https://www.ti.com/lit/ds/symlink/drv8833.pdf?ts=1745260143352&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FDRV8833%252Fpart-details%252FDRV8833PWP

Motors

N20 DC motor

- Shaft Length: 8mm-10mm
- Shaft Diameter: 2.95 mm-3.00mm (D Shape Shaft Side 2.45mm-2.5mm)
- Overall Length: 34 mm-36mm
- Overall Width/Depth: 10mm x 12mm
- LxWxH: 15.2mm-16mm x 12mm x 10mm
- Net Weight: 10g



[Pololu - 298:1 Micro Metal Gearmotor HPCB 6V with 12 CPR Encoder, Side Connector](#)

[Pololu - 298:1 Micro Metal Gearmotor HPCB 6V](#)

BATTERY

[DTP603443 Specifications by SparkFun Electronics Datasheet | DigiKey](#)

Breakdown

Microcontroller chip

1. By Bit Architecture

This refers to how much data the microcontroller processes at once.

Type	Description	Example Chips	Use Case
8-bit	Simple, low power, low cost	ATmega328 (Arduino Uno), PIC16	Basic embedded systems
16-bit	Middle ground — more capable than 8-bit	MSP430, PIC24	Industrial control, metering
32-bit	High performance, rich peripherals	STM32, ESP32, ARM Cortex-M	IoT, robotics, consumer tech

2. By Core Architecture

Different companies design or license different **CPU cores**.

Core Type	Description	Popular Vendors
ARM Cortex-M	Very common 32-bit architecture	STM, NXP, Nordic, Texas Instruments
AVR	8-bit RISC core from Atmel	Used in Arduino Uno (ATmega328)
PIC	Microchip’s proprietary architecture	PIC10/12/16/18 (8-bit), PIC24 (16-bit)
RISC-V	Open-source modern core	SiFive, Espressif, Saeed
Xtensa	Used in ESP32 series	Espressif

3. By Functionality / Feature Set

Microcontrollers are also categorized by what they're meant to **do**.

Type	Description	Example
General Purpose	All-around control and sensing tasks	STM32, ATmega328
Wireless-enabled	Built-in Wi-Fi/Bluetooth/LoRa	ESP32, nRF52, CC3200
Low-power MCUs	Ultra-low energy use for battery devices	MSP430, STM32L series
Motor Control MCUs	Designed for precision motor control	STM32F3, dsPIC33
Secure MCUs	Include encryption and secure boot	NXP LPC55S6x, Microchip CryptoAuth

4. By Vendor Ecosystem

Each company builds its own line with toolchains, dev boards, and IDEs.

Vendor	Common Families	Notes
Atmel / Microchip	AVR, SAM, PIC	Arduino-compatible, widespread
STMicroelectronics	STM32 (F0, F1, L4, H7...)	ARM Cortex-M powerhouse
Espressif	ESP8266, ESP32, ESP32-S3	Wi-Fi + Bluetooth MCU chips
NXP	LPC, i.MX	Automotive & industrial
Texas Instruments	MSP430, CC3200	Ultra-low power, wireless
Silicon Labs	EFM32, Gecko	Energy-friendly applications

Color Sensors

Devices that detect the **color composition of light**, typically through Red, Green, and Blue (RGB) channels — sometimes also Clear (C), Infrared (IR), or multispectral bands.

1. Types of Color Sensors:

Type	Description	Example Chips	Use Case
Discrete RGB	Three separate photodiodes with red, green, and blue filters	Basic LED sensors	Simple color detection
Integrated RGB	On-chip array with RGB filters, single silicon die	ISL29125, TCS3200	General-purpose color measurement
RGB + Clear (RGBC)	Adds a Clear channel for brightness/luminance measurement	TCS34725, VEML6040	Ambient light sensing, display control
RGB + IR (RGBC-IR)	Adds Infrared channel to detect or compensate for near-IR light	APDS-9960, MAX44006	Gesture control, environmental sensing
Multispectral	Detects 6 to 10+ specific wavelength bands (beyond visible RGB)	AS7262, AS7341	Agriculture, material analysis

2. Major Internal Components:

Component	Function
Photodiodes	Convert light into electrical signals, each tuned to a spectral band
Color Filter Array (CFA)	Micro-layer that allows specific wavelengths to hit respective photodiodes
IR Filter	Blocks unwanted IR light (optional, enhances visible spectrum accuracy)
Clear Channel	Measures total brightness or luminance (no filter)
Analog Front-End (AFE)	Amplifies the signal from photodiodes
Analog-to-Digital Converter (ADC)	Converts analog light level to digital data
Gain & Integration Logic	Adjusts sensitivity and sampling time
Interrupt Engine	Detects color change thresholds or flicker events
Digital Interface (I ² C/SPI)	Communicates with a microcontroller

3. Output and Data:

Output Type	Description	Typical Format
RGB Values	Intensity of red, green, blue channels	16-bit digital values
Clear (C)	Total brightness (no color filtering)	16-bit digital
Lux	Ambient light level (calculated from RGB or Clear)	Integer or float
Color Temperature	Estimates CCT (Correlated Color Temperature) in Kelvin	Float (e.g. 5000 K)
Gesture / Proximity	Used in advanced modules like APDS-9960	Boolean or event flag

4. Control Interfaces:

Interface	Description	Example Chips
I ² C	Common for most modules, easy to interface	TCS34725, APDS-9960
SPI	Less common, faster alternative	Used in high-speed sensors
Analog	Basic sensors output voltage per color	TCS3200

5. Example Use Cases:

Application	Sensor Type	Why It's Used
LED color calibration	RGBC	Measure and correct output of RGB LEDs
Ambient light detection	RGBC or RGB-IR	Adjust screen brightness in phones/tablets
Gesture control	RGB-IR + proximity	Recognize swipe motions using IR changes
Line following robot	Basic RGB	Detect surface color or contrast
Object sorting	Integrated RGB	Industrial machines sorting by color
Spectral analysis	Multispectral	Precision farming, water quality, materials ID

Accelerometers

Sensors that measure **acceleration forces** (static, like gravity or dynamic like vibration/motion) along one or more axes.

1. Types of Accelerometers:

Type	Description	Example Chips	Use Case
Capacitive	Measures change in capacitance between moving microstructures	ADXL345, MMA8451	Phones, wearables, robotics
Piezoelectric	Generates voltage when internal crystal is stressed (good for vibration)	ADXL78, 352C22	Industrial vibration sensing
Piezoresistive	Measures resistance change in response to stress	MXR9100, MEMSIC MXC	Automotive crash sensing
Triaxial (3-axis)	Measures motion along X, Y, Z axes	MPU6050, LIS3DH	Most general-purpose applications
6-DOF IMU	Combines 3-axis accelerometer + 3-axis gyroscope	MPU6050, ICM-20602	Drones, robots, motion tracking
9-DOF IMU	Adds 3-axis magnetometer (orientation sensing)	MPU9250, BNO055	Full orientation + motion tracking

2. Major Internal Components:

Component	Function
MEMS Sensor (Mass + Springs)	Mechanical microstructure that deforms in response to motion
Capacitive Plates	Detect position shift of the MEMS mass through capacitance changes
Charge Amplifier	Converts small charge changes to readable voltage
Low-pass Filter	Filters high-frequency noise from the signal
Analog-to-Digital Converter (ADC)	Converts analog signal to digital for microcontroller use
Temperature Sensor / Compensation	Corrects readings for temperature drift
I²C / SPI Interface	Provides digital communication with microcontrollers
Interrupt Generator	Signals events like tap, shake, or free-fall

3. Output and Data:

Output Format	Description	Typical Format
Raw Acceleration	Digital values for X, Y, Z axes	Signed 16-bit ints
Tap Detection	Interrupt for single/double taps	Boolean/trigger
Free-Fall Detection	Signals sudden loss of acceleration	Boolean/interrupt
Orientation	Angle estimation (when fused w/ gyro)	Roll/Pitch/Yaw

4. Control Interfaces (Digital Communication):

Interface	Notes	Example Chips
I²C	2-wire, common for microcontrollers	MPU6050, ADXL345
SPI	Faster, more reliable in noisy environments	LIS3DH, LSM6DSOX
Analog	Outputs continuous voltage (rare now)	ADXL335

5. Example Use Cases:

Application	Accelerometer Type	Why It's Used
Smartphone auto-rotate	Capacitive 3-axis	Detects orientation and tilt
Step counter (pedometer)	Triaxial digital	Detects periodic walking motion
Drone stabilization	6-DOF IMU	Combines accel + gyro for motion feedback
Fall detection (elderly care)	3-axis digital	Detects sudden changes in position
Seismic/vibration sensor	Piezoelectric analog	High bandwidth and precision for vibration
Automotive crash sensors	Piezoresistive	Very fast, can trigger airbag deployment

Magnetometers

Sensors that detect **magnetic field strength and direction**, typically used for **orientation (compass)** or **environmental magnetic field detection**.

1. Types of Magnetometers:

Type	Description	Example Chips	Use Case
Hall-Effect	Detects magnetic field via voltage change in a semiconductor	A1324, DRV5055	RPM sensing, brushless motor feedback
Magnetoresistive (AMR/GMR)	Detects changes in resistance under a magnetic field	HMC5883L, QMC5883L	Digital compasses, IMUs
Fluxgate	Highly sensitive, measures low-field changes with core-based structure	Rare in consumer sensors	Geological instruments
Triaxial (3-axis)	Measures X, Y, Z vector components of magnetic field	LIS3MDL, MPU9250, BNO055	9-DOF sensor fusion, orientation tracking

2. Major Internal Components:

Component	Function
Magnetoresistive or Hall Sensors	Convert magnetic field changes to electrical signals
Signal Conditioning Circuit	Amplifies and filters raw sensor output
Temperature Compensation Logic	Stabilizes readings against thermal drift
Axis-aligned Sensing Elements	Separate sensors for X, Y, and Z directions
Multiplexer / ADC	Converts analog signal to digital for each axis
Digital Interface (I ² C/SPI)	Communicates sensor data to host system
EEPROM / Control Registers	Stores calibration data, gain, offset settings

3. Output and Data:

Output Type	Description	Typical Format
Raw Magnetic Field	Field strength in microtesla (μT) for X, Y, Z axes	Signed 16-bit integers
Heading / Azimuth	Angle from magnetic north (after fusion or calc)	Degrees or radians
Tilt-compensated compass	Fusion of accelerometer + magnetometer	Roll/Pitch/Yaw angles
Magnetic anomaly detection	Detects large deviations in expected field	Alert or threshold

4. Control Interfaces:

Interface	Description	Example Chips
I²C	Widely used for easy microcontroller integration	HMC5883L, LIS3MDL
SPI	Faster communication, preferred in noise-prone setups	LSM303AGR
Analog (rare)	Older or special sensors provide analog voltage	A1321 (Hall-effect)

5. Example Use Cases:

Application	Sensor Type	Why It's Used
Digital compass (heading)	Triaxial magnetometer	Calculates angle to magnetic north
Orientation tracking	9-DOF IMU w/ mag	Combined with gyro/accel for orientation
Drone navigation	Magnetoresistive 3-axis	Helps maintain heading and stabilize yaw
Metal detection	Hall-effect or AMR	Detects local magnetic field disturbances
Smartphone compass	Triaxial magnetometer	For maps, AR, indoor navigation
Brushless motor feedback	Hall-effect	Detect rotor position

Motor Controllers

Electronic circuits/modules that **drive and control motors** by managing current and voltage flow.

1. Types of Motor Controllers:

Type	Description	Example Chips/Modules	Use Case
H-Bridge (Brushed DC)	Controls direction and speed of brushed DC motors	L298N, L293D, TB6612FNG	Basic DC motor control
BLDC Motor Controllers	Controls sensorless or sensed brushless motors	DRV8313, ESCs (Electronic Speed Controllers)	Drones, e-bikes, fans
Servo Motor Controllers	PWM-based position control for servo motors	PCA9685, microcontroller PWM pins	Robot arms, RC systems
Stepper Motor Drivers	Drives stepper motors with precision step and direction	A4988, DRV8825, TMC2209	CNC, 3D printers, robotics
Smart Motor Controllers	Includes PID control, feedback, and communication	RoboClaw, ODrive, VESC	High-performance robotics, robotics arms

2. Major Internal Components:

Component	Function
Power MOSFETs	Switch high current to motor windings efficiently
H-Bridge Circuit	Allows polarity reversal (for forward/reverse direction control)
PWM Controller	Regulates motor speed by adjusting pulse width
Current Sensor	Monitors current draw for safety or feedback
Gate Driver IC	Controls high-side/low-side MOSFET switching
Microcontroller (optional)	Manages feedback control, I ² C/UART/USB communication
Voltage Regulator	Powers logic components from main motor supply
Heat Sink	Dissipates heat from power components

3. Control Inputs (Interfaces):

Interface	Purpose	Example Controllers
PWM	Speed/direction control for DC/Servo motors	TB6612FNG, PCA9685
I ² C/SPI/UART	Digital control and monitoring	VEESC, RoboClaw
Analog	Potentiometer-based manual control	Simple DIY motor modules
RC Signal	For hobby ESCs and servos	Brushless ESCs (SimonK, BLHeli)

4. Example Use Cases:

Application	Motor Controller Type	Why It's Used
Line-following robot	TB6612FNG or L298N	Compact brushed DC control
3D printer	DRV8825 or TMC2209	Silent, precise stepper control
Drone	BLHeli ESC (BLDC control)	High-speed brushless control
Robotic arm	Smart controller (RoboClaw)	Feedback + high-torque coordination
RC Car	Hobby ESC + servo	Mixed brushed and PWM control

Power Management Modules

Modules that regulate, distribute, and manage **power delivery** to electronic circuits, including **voltage conversion**, **battery charging**, **protection**, and **switching**.

1. Types of Power Management Modules:

Type	Description	Example Modules	Use Case
Linear Regulators (LDO)	Step down voltage with simple circuitry, low noise	AMS1117, LM7805	Clean power for sensors, analog circuits
Buck Converter (Step-down)	Efficiently reduces voltage using switching and inductors	MP1584, LM2596	Battery-powered systems needing 3.3V or 5V
Boost Converter (Step-up)	Increases voltage from low input (e.g., 3.7V → 5V)	MT3608, Adafruit PowerBoost	LiPo battery to USB 5V output
Buck-Boost Converter	Handles inputs both above and below output voltage	TPS63000, LTC3440	Fluctuating input (e.g., solar power)
Battery Charging Modules	Safely charge Li-ion/LiPo batteries	TP4056, MCP73831	Rechargeable electronics, IoT nodes
Power Path / Load Sharing	Auto switch between USB, battery, or solar input	LiPo Rider Plus, BQ24074	Portable or solar-powered systems
Fuel Gauges / Battery Monitor	Reports battery % and health	MAX17043, BQ27441	Wearables, long-life smart devices

2. Major Internal Components:

Component	Function
Switching MOSFETs	Controls energy flow in buck/boost conversion
Inductors & Capacitors	Store and smooth energy for switching regulators
Feedback Loop Controller	Regulates voltage by adjusting duty cycle or current
Current Sensor	Measures battery charge/discharge rate
Charge Controller IC	Manages charging cycles and battery protection
Protection Circuit (OCP, OVP)	Prevents overcurrent, overvoltage, thermal faults
Battery Connector (JST-PH)	Connects LiPo/Li-ion batteries (usually 2.0mm)
Voltage Regulators (LDO or DC-DC)	Sets output voltage
Power Path Logic / ORing FET	Seamless source switching (e.g., solar → battery → USB)

3. Output and Data:

Output Type	Description	Typical Format
Regulated Voltage Output	Clean 3.3V / 5V / custom output	Analog voltage (V)
Charging Status	Indicates charging, full, or fault status	LED or digital flag
Battery Voltage	Raw battery voltage measurement	Analog or digital (ADC)
Battery % / SoC	State-of-charge for smart fuel gauges	Digital (I ² C data)
Power Source Status	Indicates USB, battery, or solar input	Digital logic / flags

4. Control Interfaces:

Interface	Purpose	Example Chips
I²C	Used in smart fuel gauges to report battery data	MAX17043, BQ27441
Analog	Simple modules use analog out for voltage reading	TP4056 + voltage divider
USB	For charging via USB-C or micro-USB	LiPo Rider Plus, MCP73831
Solar	Alternative DC input (with voltage regulation)	LiPo Rider, Adafruit Solar

5. Example Use Cases:

Application	Module Type	Why It's Used
ESP32 IoT sensor node	Boost or Buck module	Powers 3.3V ESP from 3.7V LiPo or 5V USB
Portable data logger	Power path + charger	Switches between USB charging and battery power
Solar-powered weather station	Solar + battery charger	Supports trickle charging and storage
Wearable fitness tracker	Fuel gauge + LDO	Accurate battery % + low-noise supply
Robot with 12V motors	Buck converter	Steps down 12V battery to 5V for sensors/MCU

Motors

Electromechanical devices that convert **electrical energy into mechanical rotation** — used for motion, actuation, and force generation in robotics, automation, and everyday electronics.

1. Types of Motors:

Type	Description	Example Motors	Use Case
Brushed DC Motor	Simple 2-wire motor using brushes to switch current	N20, RS-775	Wheels, fans, pumps, low-cost drives
Brushless DC (BLDC)	No brushes; uses electronic commutation for high efficiency	Drone outrunners, gimbal motors	Drones, electric vehicles, robotics
Geared DC Motor	DC motor with built-in gear reducer for torque	N20 w/ gearbox, 25GA370	Wheeled robots, automation arms
Coreless DC Motor	Lightweight, fast response motors with low inertia	Vibration motors, pager motors	Haptics, tiny robotics
Servo Motor (RC)	Motor with built-in position control using PWM input	SG90, MG996R	RC vehicles, robotic joints
Stepper Motor	Moves in precise fixed steps without feedback	28BYJ-48, NEMA17	CNC, 3D printers, automation
AC Motors	Powered by alternating current (mains or inverter-controlled)	Induction motor, synchronous motors	Industrial machinery, fans

2. Major Internal Components:

Component	Function
Rotor (Armature)	Rotating part typically has windings or magnets
Stator	Stationary part that produces a magnetic field
Commutator (brushed only)	Mechanically switches current in rotor windings
Brushes (brushed only)	Transfer power to rotating windings via contact
Magnets or Coils	Provide permanent or electromagnetism for interaction
Encoder / Feedback Sensor (optional)	Tracks position, speed (in BLDC or servo motors)
Gearbox (geared motors)	Reduces speed and increases torque
Hall-effect Sensors (BLDC)	Detect rotor position for electronic commutation

3. Output and Data:

Output Type	Description	Units / Format
Mechanical Rotation	Angular motion from electrical input	Degrees, RPM
Torque	Twisting force output	Newton-meters (N·m)
Speed (optional)	Some motors have feedback for velocity	RPM, degrees/s
Position (servo/stepper)	Controlled angular displacement	Degrees, steps

4. Control Interfaces:

Interface	Description	Used In
DC Voltage	Voltage level controls speed	Brushed DC, Coreless
PWM (Pulse Width Modulation)	Duty cycle controls speed/position	Servos, DC motors
Step + Direction Pins	Step pulses move motor, direction pin sets rotation	Stepper motors
I ² C / UART / CAN (Smart motors)	Data interface to set speed/position	Smart servos, robotic arms
RC Signal (PWM Pulse Width)	1–2ms pulse width determines angle	Hobby servos

5. Example Use Cases:

Application	Motor Type	Why It's Used
RC car steering	Servo Motor (SG90)	Simple PWM control for fixed-angle motion
Wheeled robot	Geared Brushed DC Motor	High torque at low speed with easy control
Drone propulsion	Brushless DC (BLDC)	Lightweight, efficient, fast response
3D Printer axis control	Stepper Motor	High precision, open-loop motion
Smart gripper (robot arm)	Smart Servo or BLDC + Encoder	Position feedback and torque control
Fan or blower	Brushed DC or AC Motor	Simple continuous rotation